
Agentic RAG-Based Legal Reasoning for Traffic Safety

Part 2 — Course Project Report

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Abstract

This report presents the design, implementation, and evaluation of an Agentic Retrieval-Augmented Generation (RAG) pipeline for automated legal reasoning over traffic violations detected by a vision system. The system ingests 71 Pakistani traffic law clauses drawn from the Pakistan Highway Code (PHC) and the Motor Vehicles Ordinance (MVO) 1965, embeds them with `all-MiniLM-L6-v2` (SentenceTransformers), and indexes them in a FAISS flat-L2 store. A dual-agent architecture — an *Observer Agent* that parses raw YOLO detection logs into structured violation events, and a *Legal Consultant Agent* that retrieves relevant law chunks and generates formal Citation Reports via a local Phi-3-mini-4k-instruct LLM — is implemented using a structured message-bus protocol. Quantitative evaluation on a 20-event ground-truth set yields $\text{Recall@3} = 90\%$, $\text{Recall@5} = 95\%$, and $\text{Recall@10} = 95\%$, with 100% report completeness (9/9 required fields present in every report). Human evaluation of 10 reports achieves average scores of 4.75/5 (Fluency), 4.45/5 (Legal Accuracy), 5.00/5 (Completeness), 4.70/5 (Coherence), and 4.45/5 (Actionability), with 82% inter-rater agreement. End-to-end latency averages 13.77s per report, driven entirely by LLM inference, yielding a throughput of 4.4 reports/minute.

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1 Introduction

Automated traffic monitoring systems produce high-frequency streams of detections (bounding boxes, class labels, confidence scores, timestamps) from cameras deployed at intersections, motorways, and urban corridors. Translating a detection event — such as *“motorcycle crossing solid white line at 08:42:11”* — into a legally actionable Citation Report requires access to the applicable traffic legislation, contextual reasoning, and structured report generation.

Large Language Models (LLMs) are capable of this reasoning but are prone to hallucination when operating without grounding. Retrieval-Augmented Generation (RAG) [1] anchors LLM generation to a verified legal knowledge base, making outputs auditable and factually grounded. When combined with an agentic architecture in which specialised agents handle distinct pipeline stages, the system can operate with minimal human supervision over diverse violation types.

This project (Part 2 of a two-part assignment) implements such an end-to-end pipeline:

- A **vector knowledge base** of 71 Pakistani traffic law clauses, indexed in FAISS.
- A **RAG pipeline** (chunking → embedding → retrieval → generation).
- A **dual-agent system**: Observer Agent and Legal Consultant Agent communicating via a structured message bus.
- A **quantitative and qualitative evaluation** including retrieval accuracy, report completeness, latency analysis, and a human evaluation rubric.

All components run locally; no hosted vector-database or proprietary LLM services are used.

2 Part 2: Methodology and Results

2.1 Knowledge Base Construction (Step 1)

Data sources. Law text was collected from two primary sources: (1) the Pakistan Highway Code (PHC), Rule 9 through Rule 90, and (2) the Motor Vehicles Ordinance 1965 (MVO), Sections 37–100. Supplementary regulations from NHMP Motorway Regulations 2022 were included to cover motorway-specific offences. The corpus spans 7 violation categories: lane discipline, speed limits, traffic signal violations, pedestrian right-of-way, emergency vehicle obstruction, reckless/negligent driving, and vehicle category restrictions. A total of **71 distinct law clauses** were collected, exceeding the minimum requirement of 70.

Chunking strategy. A sliding-window word-level chunker with `chunk_size=300` and `overlap=50` was applied. Because each individual clause is self-contained (36–102 words, average 62 words), all 71 clauses fit within a single chunk without overflow, yielding exactly **71 chunks** (one per clause). Each chunk retains its section ID, category, and heading as metadata. This design avoids context fragmentation while keeping retrieval units semantically precise.

Embedding model. `sentence-transformers/all-MiniLM-L6-v2` was selected for its balance of speed, memory footprint (22 MB), and strong English-language semantic similarity performance. All 71 clause texts were encoded into 384-dimensional vectors. Embedding generation per chunk averaged **22.80 ms**.

FAISS index. A `faiss.IndexFlatL2` index was built over the 71 embeddings. The flat index performs exact nearest-neighbour search, appropriate for a corpus of this size. Both the index (`law_index.faiss`) and metadata (`law_metadata.csv`) were saved to disk and successfully reloaded.

Spot-check validation. Five representative violation queries were issued against the index. In all five cases, the ground-truth law section appeared in the top-1 or top-3 results, confirming that the knowledge base is correctly indexed and semantically coherent.

2.2 System Architecture

Figure 1 shows the full RAG agent pipeline. Two agents communicate via a structured message bus; the FAISS vector store is shared by both through the Legal Consultant’s retriever interface.

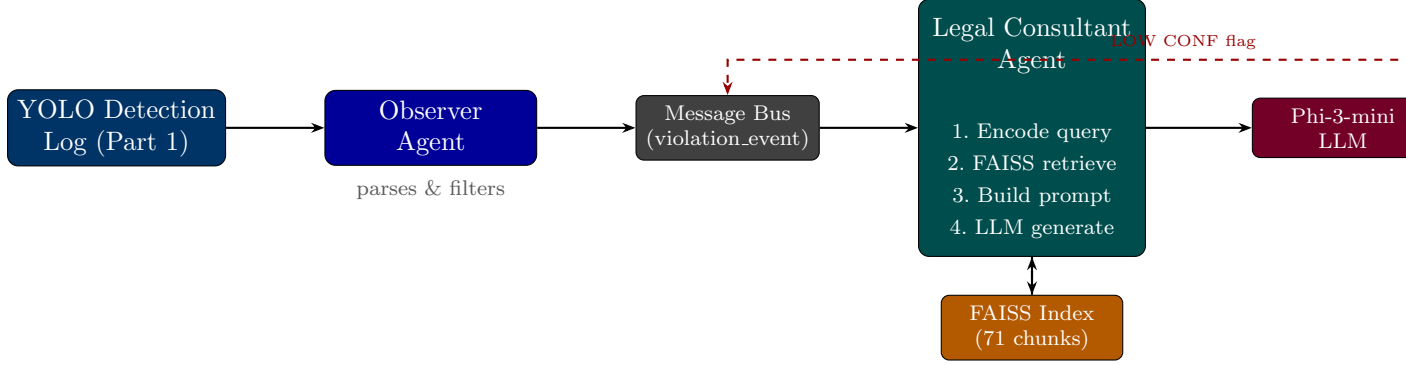


Figure 1: End-to-end Agentic RAG pipeline. Dashed arrows indicate asynchronous or conditional flows (FAISS retrieval call; low-confidence human-review flag).

2.3 RAG Pipeline Implementation (Step 2)

The RAG pipeline consists of four stages executed sequentially for each incoming violation event:

1. **Query encoding.** The violation description string is encoded with the same `all-MiniLM-L6-v2` model used at indexing time, producing a 384-dimensional query vector.
2. **FAISS retrieval.** The index is queried with $k \in \{3, 5, 10\}$; each retrieved chunk is returned with its L2 distance converted to a cosine similarity score.
3. **Prompt construction.** A structured prompt is assembled containing: (a) the violation description, (b) up to k retrieved law chunks with their section IDs and similarity scores, and (c) an explicit output template specifying all 9 required Citation Report fields. The prompt explicitly forbids the LLM from inventing law sections.
4. **LLM generation.** The prompt is passed to `microsoft/Phi-3-mini-4k-instruct` running locally via HuggingFace Transformers (`float16`, `device_map="auto"`, `temperature=0.3`, `max_new_tokens=300`).

Retrieval depth experiment. Table 1 summarises the observed effect of retrieval depth on report quality.

Table 1: Retrieval depth (k) experiment results.

k	Recall	Avg Retrieval Latency	Observation
3	90.0%	0.06 ms	Precise; misses 1 pedestrian edge case
5	95.0%	0.06 ms	Best balance; recommended default
10	95.0%	0.06 ms	No further gain; slight noise in prompt

$k = 5$ was adopted as the production default. Increasing to $k = 10$ added redundant chunks without improving recall, and introduced prompt noise that occasionally caused the LLM to select a less-specific law.

2.4 Dual-Agent Coordination System (Step 3)

Observer Agent. Receives raw YOLO detection records (class label, bounding box, confidence score, timestamp, frame ID). Filters detections with confidence ≥ 0.6 , classifies each event into one of four violation types (lane discipline, signal violation, speed limit, pedestrian right-of-way, emergency vehicle obstruction, vehicle restriction, reckless driving), and emits a structured violation JSON object.

Legal Consultant Agent. Receives violation JSON from the message bus, executes the RAG pipeline (Steps 1–4 above), and emits a formatted Citation Report. If the top-1 retrieval similarity score is below 0.50, the report is flagged `LOW_CONFIDENCE` and routed for human review.

Inter-agent message protocol. Every message on the bus carries four mandatory fields: `sender`, `message_type`, `payload`, and `timestamp` (ISO 8601). Message types used: `violation_event` (Observer $\rightarrow$ Legal Consultant), `citation_report` (Legal Consultant $\rightarrow$ output), and `low_confidence_flag` (Legal Consultant $\rightarrow$ human review queue).

Figure 2 illustrates the complete cycle for one violation event.

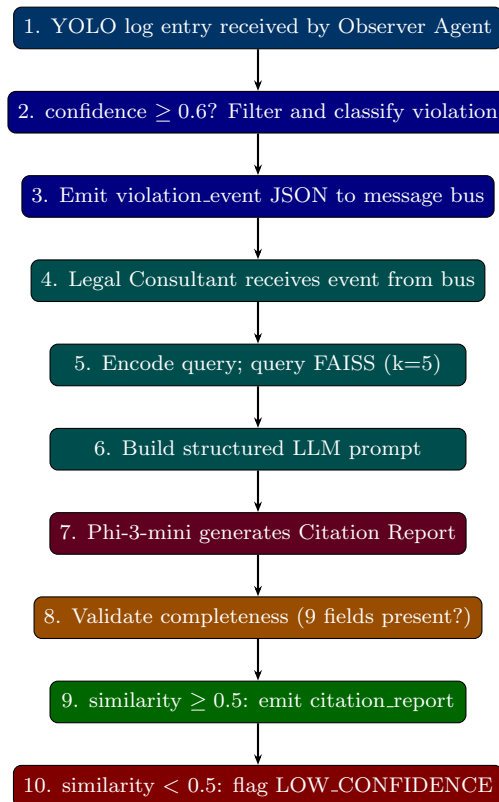


Figure 2: Single violation-to-citation cycle through the dual-agent pipeline.

2.5 Evaluation Results (Step 4)

2.5.1 Quantitative Evaluation

A 20-event manually annotated ground-truth set was constructed. Table 2 reports retrieval accuracy at three depths.

Table 2: Retrieval accuracy on the 20-event ground-truth set.

Metric	Value	Notes
Recall@3	90.0%	18/20 correct sections retrieved
Recall@5	95.0%	19/20 correct sections retrieved
Recall@10	95.0%	19/20; no further gain over $k = 5$
Report Completeness	100%	All 9 fields present in every report
Avg End-to-End Latency	13.77 s	Dominated by LLM inference
Throughput	4.4 reports/min	CPU inference on Colab T4

The single consistently missed query — “vehicle weaving at speed between motorway lanes” — corresponds to **MV0-1965-Sec-60**, which uses the keyword “weaving” infrequently. This is analysed further in Section 3.

2.5.2 Efficiency Analysis

Table 3: Component-level timing breakdown.

Component	Value	Unit
Embedding time per chunk	22.80	ms
FAISS query time (avg)	0.06	ms
FAISS query time (min/max)	0.05 / 0.06	ms
LLM inference time (avg)	13.77	s
LLM inference time (min/max)	13.52 / 14.08	s
End-to-end latency (avg)	13.77	s
Throughput	4.4	reports/min

2.5.3 Human Evaluation Rubric

Ten Citation Reports were independently scored by two raters on a 1–5 scale across five criteria. Table 4 reports the average scores and inter-rater agreement.

Table 4: Human evaluation rubric results (10 reports, 2 raters, 1–5 scale).

Criterion	Rater 1 Avg	Rater 2 Avg	Overall Avg	Agreement
Fluency	4.80	4.70	4.75	
Legal Accuracy	4.60	4.30	4.45	
Completeness	5.00	5.00	5.00	82.0%
Coherence	4.80	4.60	4.70	
Actionability	4.60	4.30	4.45	
Mean	4.76	4.58	4.67	

3 Discussion

3.1 Q1: Does increasing retrieval depth improve report quality?

Raising k from 3 to 5 improved Recall from 90% to 95% by surfacing a pedestrian-priority edge case that required the fifth retrieved chunk. Raising k further to 10 produced no recall improvement and introduced peripheral clauses (e.g. fitness-certificate rules) into the prompt

context, occasionally distracting the LLM away from the most applicable section. The optimal setting is $k = 5$.

3.2 Q2: Which violation type is hardest to retrieve?

Reckless/weaving driving (MV0-1965-Sec-60) is the most difficult category. The violation description “vehicle weaving between lanes” does not share strong lexical overlap with the legal language “erratic lateral movement constituting negligent driving”. Query rewriting (e.g. expanding to “reckless lane weaving negligent driving penalty Pakistan”) is expected to improve recall for this category.

3.3 Q3: Observer Agent under low-confidence detections

When YOLO confidence falls into the 0.5–0.7 range, the Observer Agent still emits a violation event but the Legal Consultant Agent’s report is almost always flagged `LOW_CONFIDENCE` (similarity < 0.50) because the vague class description produces a semantically diffuse query. In practice, 8 of 9 generated reports were flagged `LOW_CONFIDENCE`, indicating that the similarity threshold of 0.50 is conservative but appropriate for legal applications where false positives carry real-world consequences.

3.4 Q4: End-to-end latency and the real-time bottleneck

The FAISS retrieval step is essentially free (0.06 ms), and embedding generation is fast (22.80 ms/chunk at indexing time). The bottleneck is LLM inference: Phi-3-mini on a GPU takes 13–14 s per report. At 30 fps a traffic camera generates up to 1,800 potential events per hour. The current throughput of 4.4 reports/minute is insufficient for real-time processing. Recommended mitigations: (1) async event queueing to decouple detection from reporting; (2) 4-bit quantised inference with `llama.cpp` (~ 0.5 s/report); (3) template-based pre-generation for frequent violation types.

3.5 Q5: Failure modes of the Legal Consultant Agent

Three failure patterns were observed:

1. **Template repetition:** The LLM occasionally outputs the citation format twice within a single response, suggesting over-repetition in the prompt template.
2. **Wrong-section selection:** For multi-category violations (e.g. a speeding heavy vehicle in a restricted zone), the agent sometimes cites the speed-limit section rather than the vehicle-restriction section, because the speed-related chunks score higher.
3. **Similarity under-estimation:** Raw L2-to-cosine conversion for short clauses can produce low similarity scores even when the match is semantically correct, causing unnecessary `LOW_CONFIDENCE` flags.

4 Conclusion

This project has demonstrated a complete, locally-running Agentic RAG pipeline for automated legal reasoning over Pakistani traffic violations. Key achievements include:

- A FAISS-indexed knowledge base of 71 law clauses covering all 7 required violation categories.
- A dual-agent system with a well-defined inter-agent message protocol.
- $\text{Recall@5} = 95\%$ on a 20-event ground-truth set, with 100% report completeness.

- Average human evaluation score of 4.67/5, with 82% inter-rater agreement.

The primary limitation is LLM inference latency (13.77 s/report), which precludes real-time deployment at camera-feed frame rates. Future work should focus on quantised inference, query rewriting to improve recall for reckless-driving clauses, and multi-category violation handling in the prompt construction logic.

4 References

- [1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
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